

Johnny Clem

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Video Compression & Encoding · Low-Latency Streaming · C++ / Swift · ffmpeg · OpenCV · AVFoundation

Staff-level engineer and former CTO of the team behind FiLMiC Pro — one of the most advanced professional video capture applications ever built for iOS. Named inventor on a US patent for real-time video production protocols. Career spans custom H.264 encoder design, low-latency streaming systems, C++ media SDKs, computer vision integration, and twenty years of ffmpeg fluency rooted in a pre-engineering career as a professional video editor and on-set DIT.

EXPERIENCE

CTO / Lead iOS Engineer — Cinegenix (FiLMiC Pro)

2014–2021 · Seattle, WA

- Designed and shipped a **custom H.264 encoder** for FiLMiC Remote: hardware-accelerated encoding of product-optimized NALUs via VideoToolbox compression sessions, bypassing AVFoundation's standard recording pipeline for lowest achievable encode-to-network latency
- Built a **second custom H.264 encoder** for the core FiLMiC Pro recording path, exposing fine-grained control over SPS/PPS emission, IDR interval, slice sizing, and rate control parameters unavailable through high-level AVFoundation APIs
- Integrated **OpenCV** into the real-time capture pipeline for on-device frame analysis and image processing tasks beyond native AVFoundation capabilities
- Designed and shipped the **C++ audio capture and processing engine**: real-time metering, monitoring, gain control, multi-channel routing, and synchronized A/V recording
- Built a proof-of-concept **ProRes software encoder** for iPhone (~5–6 fps ProRes Proxy/LT), demonstrating the codec performance gap later closed when Apple shipped hardware ProRes on A15+
- Led the ground-up rewrite of FiLMiC Pro in Swift 2.0 (iOS 9): full AVFoundation capture pipeline, manual exposure/focus/WB control, LOG/flat color profiles, configurable H.264/HEVC codec parameters. #1 video app on the App Store for multiple years; used on feature films and broadcast journalism

iOS Engineer — Paladin Innovators

2012–2014 · San Diego, CA

- Designed and shipped a **low-latency live video streaming system**: iPhone to production video switcher over point-to-point Wi-Fi, achieving **sub-100ms glass-to-glass latency** at 720p60 on 2012-era hardware
- Built custom NALU packaging to minimize decode-side buffering; tuned hardware-accelerated H.264 encoding (B-frame elimination, GOP structure, rate control) for real-time delivery
- Used **ffmpeg** extensively on the server/switcher side for transcoding, muxing, and stream analysis throughout development and production deployment
- Production-deployed at live multi-camera events including broadcasts and corporate productions

Senior iOS Engineer / SDK Tech Lead — Truepic

2023–2024 · San Diego, CA (Remote)

- Tech lead on Truepic's image capture, sign, and validate SDK for iOS — a cryptographic content provenance system verifying photos/videos are unmanipulated from moment of capture
- SDK built entirely on underlying **C++ libraries** that I interoperated with and worked on daily: image capture pipeline, cryptographic signing, and validation flows
- Deep AVCaptureSession integration to embed provenance metadata before any post-processing — pipeline design with direct parallels to content integrity and watermarking in streaming delivery

Director of Digital — Belief Agency

2021–2022

- Led digital strategy and engineering for a creative agency, bridging product vision with technical execution

Senior iOS Engineer — Realtor.com (Move/RDC)

2025–Present · Austin, TX (Remote)

- Staff-level iOS engineering; hardened CI security and reduced pipeline build times by 33%

PATENT

US20150350041A1 — Protocols & Mechanisms of Communication Between Live Production Server and Mobile or Remote Clients

Inventor: Jonathan Clem, Dean Bisogno · Assignee: Paladin Innovators · Filed: May 30, 2014

WebSocket-based command/response protocol for coordinating real-time video production between a centralized encoding server and mobile thin clients. Covers session management, encode state control, multi-client synchronization, and hardware routing over a unified protocol.

PRE-ENGINEERING: VIDEO EDITOR & DIT

Professional video editor and on-set Digital Imaging Technician on film and broadcast productions. Built custom compression profiles in **ffmpeg** and Apple Compressor to maximize perceptual quality under extreme bitrate constraints — including squeezing a weekly hour-long production into H.264 conforming to iPod Video spec (baseline profile, ~1.5 Mbps). Fluent in ffmpeg since the early 2000s.

TECHNICAL EXPERTISE

Video & Codec: Custom H.264 encoder design, NALU-level bitstream control, VideoToolbox, hardware-accelerated encoding/decoding, H.264/AVC, HEVC, ProRes, rate control, GOP tuning, low-latency streaming

C++ & Systems: Production C++ in audio/video engines and image capture SDKs (FiLMiC Pro, Truepic); real-time performance-critical media pipelines

Image & Signal Processing: OpenCV integration, real-time frame analysis, color space management, ISP pipeline awareness, image provenance/watermarking

Tools & Frameworks: ffmpeg (20 years), AVFoundation, AVCaptureSession, Swift, Objective-C, Python

iOS Platform: Swift, SwiftUI, UIKit, Combine, modular architecture, large-scale iOS systems, CI/CD

SELECTED PRODUCTS

FiLMiC Pro — Industry-leading professional video capture, #1 App Store video app · **FiLMiC Remote** — Real-time streaming camera monitor/controller · **DoubleTake** — Simultaneous multi-camera recording · **CinePro** — #1 App Store video app (2012) · **Truepic SDK** — Cryptographic image provenance

Improves encoding systems, not just features. Brings twenty years of hands-on compression intuition from both sides of the camera. Leads through technical depth and trust.